

- 1 The photograph shows flowers of a water hyacinth plant, *Eichhornia crassipes*.



(Source: © Natalia Kuzmina / Alamy Stock Photo)

These plants are an invasive species found in lakes and rivers throughout Asia. They grow rapidly, producing many flowers and seeds.

A student observed that each flower has six anthers and one stigma.

Each anther was observed to release many pollen grains.

When pollen grains from the water hyacinth are placed in a sucrose solution they grow pollen tubes.

These could be seen growing within 30 minutes.

The student wanted to find out if an extract from the stigma stimulated the growth of pollen tubes.

(a) Describe an experiment to investigate the effect of this extract on the rate of pollen tube growth.

(6)

(b) The tip of the pollen tube is protected by an outer layer of pectin.

Pectin is a polysaccharide.

Starch grains, mitochondria and vesicles are present in the tip of the pollen tube.

Describe how these structures are used in the production of pectin at the tip of a pollen tube.

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(Total for Question 1 = 10 marks)